

Point Vortex Dynamics on Closed Surfaces: A DEC Formulation and a Hidden Jacobian Invariant

RUDRESH VEERKHARE and ALBERT CHERN

Point vortices are one of the oldest models in fluid dynamics, but on curved surfaces they still hide structure that is invisible in the classical planar story, especially on higher-genus surfaces. In this report we revisit point vortex dynamics on closed surfaces of arbitrary genus through the lens of discrete exterior calculus (DEC), cohomology, and complex analysis. On the geometric side, we build a DEC-based point-vortex solver on arbitrary closed meshes that treats the $2g$ global handle circulations as explicit degrees of freedom coupled to the vortex motion. Tracking these circulations in the spirit of Kelvin’s theorem leads to a simple real conservation law: a $2g$ -dimensional invariant that links vortex positions to how the flow winds around the handles. On the complex-analytic side, we show how this same invariant naturally extends the classical complex-potential formulation from the plane to compact Riemann surfaces and line bundles, where it lives as a nontrivial element of the Jacobian variety. In short, a cohomological correction needed to simulate vortices on arbitrary meshes turns out to define a genuinely geometric invariant in the algebraic topology of the underlying surface.

1 BIG PICTURE

Two-dimensional incompressible hydrodynamics has been a playground for both physics and mathematics for more than a century. In the vorticity formulation of incompressible Euler, incompressibility appears as a divergence-free constraint on the velocity, while vorticity measures the local “swirl” of the flow. On simply connected planar domains one introduces a scalar stream function ψ whose Laplacian equals the vorticity and recovers the velocity as a 90° rotation of $\nabla\psi$. Already in the 19th century, Helmholtz, Kirchhoff and others realized that a particularly simple yet dynamically rich toy model is obtained by concentrating vorticity at isolated points, *point vortices*, thought of as idealized tornadoes or whirlpools [Helmholtz 1858; Kirchhoff 1876; Aref 2007; Newton 2001]. In the plane this story is beautifully clean: the velocity is generated by a single complex-valued potential, the point vortices follow a Hamiltonian ODE, and classical complex analysis (harmonic functions, conformal maps, meromorphic functions) provides closed-form descriptions of many configurations. In that sense the “point vortex in the plane” problem is a largely classical chapter of 19th- and early-20th-century mathematics.

Once we move off the plane onto a closed surface, two new ingredients appear: *curvature* and *topology*. Curvature already matters on the sphere: work of Kimura, Polvani–Dritschel, Dritschel–Boatto, and others shows that even a single vortex can drift under its own “image” vorticity once the global constraints on the Green’s function are enforced, and stability properties can change purely due to curvature [Kimura and Okamoto 1987; Kimura 1999; Polvani and Dritschel 1993; Dritschel and Boatto 2015]. On surfaces with handles, there is an additional, genuinely topological phenomenon: the fluid can carry persistent global swirls around non-contractible loops. In modern language these are the harmonic 1-form components in the Hodge decomposition of the velocity field, and physically they are circulations wrapping around handles that cannot be represented by any choice of stream function. Recent work by Gustafsson and coauthors formulates point vortex dynamics on closed Riemannian

surfaces and makes these harmonic degrees of freedom explicit dynamical variables, coupled to the vortex positions by Kelvin’s circulation theorem [Gustafsson 2019, 2022; Grotta-Ragazzo 2017; Grotta-Ragazzo et al. 2024]. The phase space is no longer “just” vortex positions: it also includes a finite-dimensional space of global circulations constrained by the topology of the surface.

Historically, the theory has been developed from several angles. There is the classical planar Kirchhoff–Routh formulation and the Hamiltonian N -vortex problem. There are complex-analysis treatments on simply connected or multiply connected planar domains via conformal mapping, where the flow is encoded by a single-valued or multivalued complex potential. There are intrinsic formulations on constant-curvature or closed Riemannian surfaces, where curvature and topology enter through Green/Robin functions and global circulations [Helmholtz 1858; Lin 1941; Saffman 1992; Newton 2001; Aref 2007; Crowdy and Marshall 2005; Crowdy 2005, 2007; Sakajo and Shimizu 2016; Bogomolov 1977; Kimura 1987; Kidambi and Newton 1998; Boatto and Koiller 2008, 2015; Dritschel and Boatto 2015; Gustafsson 2019, 2022; Grotta-Ragazzo 2017; Grotta-Ragazzo et al. 2024]. These smooth formulations are powerful but geometry-specific: every new metric or topology tends to require its own analytic construction (Green’s functions, conformal models, Robin functions) and the harmonic handle sector is often encoded implicitly inside those objects.

On the numerical side, there is a long CFD tradition of vorticity- and vortex-based methods. Classical vortex methods treat vorticity as particles or blobs and integrate them with particle–grid schemes, variants of Biot–Savart summation, and sophisticated time integration [Cottet and Koumoutsakos 2000; Chorin 1994; Leonard 1980; Majda and Bertozzi 2001]. These methods are usually designed for planar or simple geometries and do not expose the topological degrees of freedom on higher-genus surfaces as an explicit finite-dimensional cohomology sector [Chorin 1994; Pozrikidis 2011; Boffetta and Ecke 2012]. Discrete Exterior Calculus (DEC) and related discrete differential-forms methods were developed precisely to put calculus on simplicial meshes in a structure-preserving way [Hirani 2003; Desbrun et al. 2008; Gillette 2009]. They have been used to build fluid solvers such as *Stable, Circulation-Preserving, Simplicial Fluids* and DEC-based surface Navier–Stokes schemes that work directly on triangle meshes, including closed surfaces [Elcott et al. 2007; Camporesi and Kanso 2016]. These schemes already expose some topological structure, but they are typically formulated for full-field evolution rather than reduced point-vortex dynamics with explicit harmonic degrees of freedom.

More recently, the *Fluid Cohomology* framework [Yin et al. 2023b,a] made the topological component completely explicit: on multiply connected domains and higher-genus surfaces there are additional harmonic/cohomology degrees of freedom that previous solvers either froze or ignored, and there are missing equations of motion for these components. Fluid Cohomology derives a fully topology-aware

formulation in which these harmonic variables evolve on the same footing as the usual vorticity. Our project extends this geometric/cohomological picture to the point-vortex setting on arbitrary closed, triangulated surfaces, using DEC as the discretization framework.

The surprising twist and the main conceptual revelation of this note, is that once we insist on treating the harmonic handle component as honest, finite-dimensional dynamical data, a very rigid geometric invariant pops out. At the level of differential forms, we show that a certain $2g$ -dimensional real quantity that combines vortex positions and handle circulations is conserved under the incompressible Euler evolution. When we equip the surface with a compatible complex structure and reinterpret the same data in the language of Riemann surfaces, holomorphic 1-forms, and line bundles, this real invariant naturally repackages as a point in the Jacobian variety $\text{Jac}(M)$, the complex torus that classifies degree-zero holomorphic line bundles. In other words, the PDE/DEC dynamics of point vortices and harmonic circulations appears to preserve a nontrivial algebraic-geometric object: the isomorphism class of a certain line bundle associated to the flow. This observation motivates us to extend the classical complex-analytic picture of point vortices from simply connected planar domains to general compact Riemann surfaces.

Conceptually, this project lives at the intersection of these stories:

- **Explicit harmonic degrees of freedom on meshes.** We formulate incompressible Euler and point vortex dynamics on closed surfaces in vorticity–streamfunction form, make the $2g$ -dimensional harmonic component explicit using Hodge theory, and discretize it using DEC so that the same structure survives on arbitrary triangulated surfaces.
- **A DEC-based point vortex solver on arbitrary genus.** We construct and implement a point-vortex solver on triangle meshes that couples a stream-function Poisson solve to an explicit discrete harmonic subspace, so that both singular vorticity and handle circulations evolve on meshes of any genus.
- **A geometric invariant that lands in the Jacobian.** By tracking harmonic circulations and vortex positions in parallel (in the spirit of Kelvin’s theorem), we derive a conserved real quantity valued in $\mathbb{R}^{2g}$. After choosing a complex structure, this invariant can be reinterpreted as a point in the Jacobian variety $\text{Jac}(M)$, suggesting that the Euler evolution preserves the isomorphism class of a certain degree-zero line bundle. At present this Jacobian-valued conservation law is established at a formal level; making the complex-analytic side fully rigorous is ongoing work.
- **A generalized complex-analysis picture using line bundles.** We revisit the classical complex-potential formulation of point vortices, which traditionally assumes a simply connected domain and a single-valued complex potential, and extend this picture to multiply connected and higher-genus surfaces by interpreting the potential as a meromorphic section of a complex line bundle over a compact Riemann surface. In this language, the apparent “multivaluedness” of the potential and the handle circulations are naturally encoded by the Abel–Jacobi map and by points of the Jacobian.

The rest of the note is organized as follows. Section 2 develops the DEC-based vorticity–streamfunction formulation, the point-vortex specialization, and the discrete harmonic basis. Section 3 adds a complex structure, reviews the relevant Riemann-surface and line-bundle language, and states a formal Jacobian-valued invariant. An appendix collects the more systematic background on Hodge theory, Picard/Jacobian, and Chern–Weil theory, and the main text focus more on intuition and the geometry of the invariant.

2 DISCRETE EXTERIOR CALCULUS FORMULATION

Our discrete formulation follows the same philosophy as Fluid Cohomology [Yin et al. 2023b,a]: we write incompressible Euler in vorticity streamfunction form on a closed surface, keep track of all cohomological degrees of freedom, and then discretize using discrete exterior calculus (DEC) so that this structure survives on a triangulated mesh [Hirani 2003; Desbrun et al. 2008]. In particular, DEC lets us work directly on arbitrary triangulated surfaces, of any curvature and genus, while preserving Stokes’ theorem and Hodge decomposition in a weak sense. The resulting discretization is *geometry-aware* and *topology-aware*: we never have to flatten the surface or precompute global Green/Robin kernels, and the extra degrees of freedom coming from handles appear automatically as a finite-dimensional harmonic subspace.

2.1 Vorticity–streamfunction on a closed surface

Let (M, g) be a closed, oriented Riemannian surface with area form dA . Let v be the fluid velocity, and let $\eta := v^\flat$ be its metric dual 1-form. The incompressible Euler equations on M can be written in the usual vector-calculus notation as

$$\partial_t v + (v \cdot \nabla)v = -\nabla p, \quad (1)$$

$$\nabla \cdot v = 0, \quad (2)$$

where p is the pressure enforcing incompressibility.

The vorticity is a scalar field ω defined by

$$\omega dA = \Omega := d\eta,$$

i.e., Ω is the vorticity 2-form. On a *simply connected* domain, incompressibility implies that the velocity is a 90° rotation of $\nabla\psi$ for some scalar stream function ψ . On a closed surface with handles, this picture is almost right but misses extra “handle” degrees of freedom; Hodge decomposition is the right language to see what is missing.

We first rewrite Euler in exterior-calculus notation:

Proposition 2.1 (Exterior-calculus form of incompressible Euler). *On a closed oriented surface M , the incompressible Euler equations (1)–(2) are equivalent to*

$$\partial_t \eta + \iota_v \Omega + dp = 0, \quad (3)$$

$$\delta \eta = 0, \quad (4)$$

where $\Omega = d\eta$ is the vorticity 2-form, δ is the codifferential, and ι_v is interior product with v .

The proof is standard “vector-calculus $\leftrightarrow$ differential-forms” book-keeping and is included in Appendix B.1 for completeness.

Remark 2.1 (Gradient part and pressure gauge on a closed surface). *Hodge decomposition splits any 1-form η into an exact part $d\phi$, a co-exact part $\star d\psi$, and a harmonic part h . On a closed surface,*

incompressibility ($\delta\eta = 0$) forces ϕ to be harmonic, hence constant, so $d\phi = 0$; the gradient mode really does disappear. Physically, one can think of the pressure as eating the gradient part, leaving only the divergence-free piece $\star d\psi + h$. We spell this out more systematically below.

2.2 Hodge decomposition and harmonic handle degrees of freedom

The missing degrees of freedom on a higher-genus surface are exactly the harmonic part in the Hodge decomposition of 1-forms.

Proposition 2.2 (Hodge decomposition for 1-forms). *Let (M, g) be a closed, oriented Riemannian surface. Then every smooth 1-form α admits an L^2 -orthogonal decomposition*

$$\alpha = d\phi \oplus \star d\psi \oplus h,$$

where ϕ, ψ are scalar functions (defined up to constants) and h is a harmonic 1-form, i.e., $dh = 0$ and $\delta h = 0$, with δ the codifferential. Moreover,

$$\dim \mathcal{H}^1(M) = \dim H^1(M; \mathbb{R}) = 2g,$$

where g is the genus of M and $\mathcal{H}^1(M)$ is the space of harmonic 1-forms.

A proof can be found in any Hodge-theory textbook; we sketch the ingredients we need in Appendix B.2.

Physically, one can read the three pieces as:

- $d\phi$: gradient (potential) component, invisible to vorticity; - $\star d\psi$: the usual stream-function component; - h : the “purely topological” part, carrying circulations that go around handles and cannot be written as gradients or curls of a single-valued scalar.

In incompressible flow, the gradient part is killed by the incompressibility constraint and the pressure gauge, so only the last two pieces survive.

Lemma 2.1 (Stream function plus harmonic handle circulations). *If η is the velocity 1-form of an incompressible flow on M , then*

$$\eta = \star d\psi + h$$

for some stream function ψ and some harmonic 1-form h . The co-exact part $\star d\psi$ is entirely determined by the vorticity $\Omega = d\eta$, while h encodes the $2g$ independent handle circulations

$$\Gamma_Y = \oint_Y \eta,$$

one for each generator γ of $H_1(M; \mathbb{Z})$ (for instance, $2g$ loops around the handles).

The short argument is: apply Proposition 2.2, impose $\delta\eta = 0$, and note that exact 1-forms integrate to zero on closed loops, so periods only “see” the co-exact and harmonic parts. Details are in Appendix B.3.

The stream-function piece is recovered from the vorticity via a scalar Poisson problem:

Lemma 2.2 (Stream-function Poisson problem on a closed surface). *Let $\Omega = \omega \, dA$ be a vorticity 2-form on a closed surface M with mean-zero scalar vorticity*

$$\int_M \omega \, dA = 0.$$

Then there exists a stream function ψ , unique up to an additive constant, such that

$$\eta = \star d\psi + h, \quad \Delta\psi = \omega,$$

where $\Delta = \delta d$ is the Laplace–Beltrami operator on scalars and h is harmonic. If the total vorticity is nonzero, one must add a constant background vorticity (or equivalent rigid-body rotation) to satisfy the global compatibility condition.

Remark 2.2 (Green’s and Robin functions vs direct Poisson solves). Classically, one solves Lemma 2.2 using the Green’s function $G(x, y)$ of the Laplace–Beltrami operator,

$$\Delta_x G(x, y) = \delta_y(x) - \frac{1}{\text{Area}(M)},$$

and writes $\psi(x)$ as a sum of Green’s functions for each vortex, plus a harmonic “Robin” term to take care of self-interaction and the mean-zero constraint. In our DEC formulation we never explicitly build G or its Robin part; we directly solve the discrete Poisson equation on the mesh, which is both simpler and more flexible for arbitrary geometries.

On a simply connected planar domain (or a genus-zero surface with boundary), the harmonic 1-form space vanishes, so $\eta = \star d\psi$ and everything is controlled by the single scalar ψ . On higher-genus closed surfaces, the harmonic subspace has dimension $2g$, and these extra degrees of freedom show up as independent handle circulations. Later, when we put a complex structure on M , this harmonic sector will be the real shadow of the Jacobian of the Riemann surface.

2.3 Point vortices and the stream function

In the point-vortex model, the vorticity is concentrated at finitely many moving points $D_a(t) \in M$ with circulations (strengths) Γ_a . The vorticity 2-form is

$$\Omega(t) = \sum_{a=1}^N \Gamma_a \delta_{D_a(t)}, \quad (5)$$

where each δ_{D_a} is a Dirac 2-current at D_a . The stream function ψ solves the scalar Poisson problem

$$\Delta\psi = \omega = \sum_{a=1}^N \Gamma_a \left(\delta_{D_a} - \frac{1}{\text{Area}(M)} \right), \quad (6)$$

with a gauge condition (for instance, zero mean or fixing ψ at one point) to remove the additive constant in the kernel of Δ .

The velocity is then

$$\eta = \star d\psi + h, \quad v = \eta^\sharp, \quad (7)$$

i.e., v is the 90° rotation of $\nabla\psi$ plus the harmonic handle circulations. The point vortices are simply advected by this velocity field:

$$\dot{D}_a(t) = v(D_a(t)). \quad (8)$$

On a planar simply connected domain, one typically packages the stream function ψ with a harmonic conjugate velocity potential ϕ into a single complex-valued *complex potential*

$$W = \psi + i\phi,$$

whose derivative encodes the velocity and whose logarithmic singularities encode the vortices. On higher-genus closed surfaces the same idea works only locally; globally there are topological obstructions coming from the harmonic part h . In the main text

we mostly stay with the real picture (6)–(8); in Section 3 we revisit the complex-potential story and interpret it using line bundles and Abel–Jacobi theory.

2.4 DEC discretization on a triangulated surface

We now discretize the above continuous picture using discrete exterior calculus. Let K be a triangulation of M with vertex set K^0 , edge set K^1 , and face set K^2 . We denote by $C^k(K)$ the space of real-valued k -cochains, i.e., functions on K^k .

The DEC ingredients are:

- Discrete exterior derivatives $d_0 : C^0 \rightarrow C^1$ and $d_1 : C^1 \rightarrow C^2$, given by oriented incidence matrices.
- Discrete Hodge stars $\star_0 : C^0 \rightarrow C^{2,*}$, $\star_1 : C^1 \rightarrow C^{1,*}$, $\star_2 : C^2 \rightarrow C^{0,*}$ mapping primal cochains to dual cochains. We use the standard cotangent-weight construction, which coincides with the circumcentric-dual DEC Hodge star on Delaunay meshes and extends naturally to general triangle meshes.
- The scalar cotangent Laplacian

$$L := d_0^\top \star_1 d_0 : C^0(K) \rightarrow C^{0,*}(K).$$

Under the inner product $\langle \phi, \psi \rangle_{L^2} := \phi^\top \star_0 \psi$ this operator is symmetric positive semidefinite. If we identify $C^{0,*}$ with C^0 via $\star_0$, this corresponds to the usual DEC Laplace–Beltrami operator

$$\Delta_0 = \star_0^{-1} L = \star_0^{-1} d_0^\top \star_1 d_0.$$

In implementation we work directly with the SPD matrix L .

We store the stream function as a scalar cochain $\psi_h \in C^0(K)$ at vertices. The discrete velocity 1-form is then

$$u_h = \star_1 d_0 \psi_h + h_h \in C^1(K),$$

where h_h is the (time-dependent) discrete harmonic 1-cochain encoding the handle circulations.

Instead of first building a primal vorticity 2-cochain and then applying $\star_2$, we assemble the dual vorticity directly on vertices.

Definition 2.1 (Discrete vorticity of point vortices). *Given point vortices at positions x_i with strengths Γ_i , we define a dual vorticity cochain*

$$\widehat{\omega}_h \in C^{0,*}(K)$$

as follows. For each vortex x_i lying in a face $f = [v_1, v_2, v_3]$ with barycentric coordinates $(\lambda_1, \lambda_2, \lambda_3)$, we distribute its strength to the three incident vertices by

$$\widehat{\omega}_h(v_j) += \Gamma_i \lambda_j, \quad j = 1, 2, 3.$$

Summing over all vortices yields a dual vorticity cochain $\widehat{\omega}_h$ that can be interpreted as the vorticity (per dual cell) associated to the Poincaré dual cells around the vertices. Conceptually, this $\widehat{\omega}_h$ approximates $\star_2 \Omega_h$, but we never explicitly form the primal 2-cochain Ω_h .

The discrete stream function is obtained from a discrete Poisson problem with right-hand side given by this dual vorticity.

Definition 2.2 (Discrete stream-function solve). *Let $\widehat{\omega}_h \in C^{0,*}(K)$ be the dual vorticity cochain assembled from the point-vortex configuration. To enforce compatibility with the constant nullspace of L ,*

we impose a Dirichlet pin at one vertex. The discrete stream function $\psi_h \in C^0(K)$ is then defined as the solution of

$$L \psi_h = \widehat{\omega}_h, \tag{9}$$

where $L = d_0^\top \star_1 d_0$. In practice we fix ψ_h at a single vertex (removing the nullspace explicitly).

Once ψ_h is known, we obtain the discrete velocity 1-cochain $u_h = \star_1 d_0 \psi_h + h_h$ and recover a piecewise linear velocity field on the mesh. Evaluating this field at the vortex positions (e.g., via barycentric interpolation on the enclosing triangle) gives the right-hand side of the point-vortex ODE (8).

2.5 Discrete harmonic basis and circulations

DEC provides a discrete analogue of Hodge decomposition:

$$C^1(K) = \text{im}(d_0) \oplus \text{im}(\delta_2) \oplus \mathcal{H}_h^1(K),$$

where $\delta_2 := \star_1^{-1} d_1^\top \star_2$ is the codifferential on 2-cochains and $\mathcal{H}_h^1(K)$ is the space of discrete harmonic 1-cochains, satisfying

$$d_1 h_h = 0, \quad \delta_1 h_h = 0.$$

For a triangulated closed genus- g surface we have $\dim \mathcal{H}_h^1(K) = 2g$, mirroring the smooth case.

We construct a discrete harmonic basis that is orthonormal with respect to the discrete L^2 inner product

$$\langle \alpha, \beta \rangle_{L^2} := \alpha^\top \star_1 \beta.$$

Lemma 2.3 (Construction of a discrete harmonic basis). *Let K be a triangulated closed genus- g surface. Then there exists a set of $2g$ discrete harmonic 1-cochains $\{\eta_a\}_{a=1}^{2g} \subset \mathcal{H}_h^1(K)$ with the following properties:*

- (1) *Each η_a is closed and co-closed: $d_1 \eta_a = 0$ and $\delta_1 \eta_a = 0$.*
- (2) *The η_a are L^2 -orthonormal:*

$$\langle \eta_a, \eta_b \rangle_{L^2} = \delta_{ab}.$$

- (3) *They form a basis: any $h_h \in \mathcal{H}_h^1(K)$ can be written uniquely as*

$$h_h = \sum_{a=1}^{2g} c_a \eta_a,$$

for some coefficients $c_a \in \mathbb{R}$.

In practice we obtain these η_a by building 1-cochains that represent fundamental cycles from a tree–cotree decomposition and then orthogonally projecting them onto the kernel of the discrete 1-form Laplacian $\Delta_1 := \delta_2 d_1 + d_0 \delta_1$; see Appendix B.4 for details.

Definition 2.3 (Discrete harmonic coordinates). *Given the basis $\{\eta_a\}_{a=1}^{2g}$ from Lemma 2.3, we write the harmonic part of the discrete velocity as*

$$h_h(t) = \sum_{a=1}^{2g} c_a(t) \eta_a.$$

The vector $c(t) = (c_1(t), \dots, c_{2g}(t)) \in \mathbb{R}^{2g}$ is called the harmonic coordinate of the flow. Its components are linearly equivalent to the circulations along a choice of homology basis on the mesh.

2.6 Evolution of harmonic circulations and a real invariant

We now explain, at a continuous level, how the harmonic coordinates evolve under the Euler flow, and how they combine with vortex positions to give a conserved quantity. The discrete version mirrors the same structure.

Let $\{h_i\}_{i=1}^{2g}$ be an L^2 -orthonormal basis of harmonic 1-forms on (M, g) ,

$$\int_M h_i \wedge \star h_j = \delta_{ij},$$

and let v be the fluid velocity with velocity 1-form $\eta := v^\flat$. Define the i -th harmonic coefficient of the flow by

$$c_i(t) := \int_M \eta(t) \wedge \star h_i. \quad (10)$$

For point vortices with strengths Γ_a at positions $D_a(t) \in M$, we write the vorticity as

$$\Omega := d\eta = \sum_a \Gamma_a \delta_{D_a(t)}, \quad (11)$$

where $\delta_{D_a(t)}$ are Dirac 2-currents at the vortex positions (we absorb the usual 2π factors into the Γ_a for simplicity).

Lemma 2.4 (Evolution of harmonic coefficients (formal calculation)). *Let (M, g) be a closed, oriented surface, and let v solve incompressible Euler with point-vortex vorticity (11), with the vortices advected by the flow, $\dot{D}_a(t) = v(D_a(t))$. For each harmonic basis form h_i choose local scalar potentials A_i near each vortex such that $\star h_i = dA_i$. Then, at the level of formal calculations with distributions,*

$$\dot{c}_i(t) = \frac{d}{dt} \left(\sum_a \Gamma_a A_i(D_a(t)) \right), \quad (12)$$

and hence the quantity

$$F_i(t) := c_i(t) - \sum_a \Gamma_a A_i(D_a(t)) \quad (13)$$

is constant in time.

Concretely, one obtains

$$\begin{aligned} \dot{c}_i(t) &= - \sum_a \Gamma_a \int_M \iota_{v(D_a(t))} \delta_{D_a(t)} \wedge \star h_i \\ &= \sum_a \Gamma_a \nabla A_i(D_a(t)) \cdot \dot{D}_a(t) \\ &= \frac{d}{dt} \left(\sum_a \Gamma_a A_i(D_a(t)) \right). \end{aligned} \quad (14)$$

which is the formula stated in Lemma 2.4. The appendix spells this out using the language of currents.

Corollary 2.1 (Real Jacobian-type invariant). *Collect the $F_i(t)$ into a vector*

$$F(t) := (F_1(t), \dots, F_{2g}(t)) \in \mathbb{R}^{2g}.$$

Then $F(t)$ is constant in time for the point-vortex dynamics. In other words, the combination of harmonic circulation coordinates $c_i(t)$ and vortex-position-dependent potentials $A_i(D_a(t))$ defines a $2g$ -dimensional real invariant of the flow.

In the DEC setting we use the intermediate formulation from (14) where $c_i(t) = \sum_a \Gamma_a \nabla A_i(D_a(t)) \cdot \dot{D}_a(t)$ and using

$$\star h_i = dA_i \implies \nabla A_i = (\star h_i)^\sharp = J h_i^\sharp, \quad \dot{D}_a(t) = v(D_a(t))$$

where $h_i^\sharp$ is the metric dual vector field and J is the 90° rotation on tangent vectors, the formal calculation in Lemma 2.4 can be rewritten as

$$\dot{c}_i(t) = \sum_a \Gamma_a v(D_a(t)) \cdot J h_i^\sharp(D_a(t)).$$

In our implementation, we approximate v and $h_i^\sharp$ using DEC (stream-function solves plus a discrete harmonic basis), evaluate the right-hand side at the vortex positions by barycentric interpolation on the mesh, and integrate this ODE system for $\{c_i(t)\}$ together with the point-vortex ODE (8) using a standard Runge–Kutta scheme (RK4). This keeps the structure of the ODE continuous and we don't need to worry about choosing branch cuts for globally multi-valued potential A_i . We give further implementation details in Appendix B.6.

3 RIEMANN SURFACES, LINE BUNDLES, AND ABEL–JACOBI THEORY

So far our discussion has been purely Riemannian: we used the metric g and Hodge theory to expose the harmonic handle degrees of freedom and built a DEC-based point-vortex formulation that respects this structure. In this section we switch on a compatible complex structure and reinterpret the same data in the language of Riemann surfaces, Jacobians, and (degree-zero) line bundles.

The goal is not to give a full algebraic-geometry treatment, but to explain how the *real* invariant from Corollary 2.1 naturally repackages into an element of the Jacobian variety $\text{Jac}(M)$, and how this element can be interpreted as the isomorphism class of a holomorphic line bundle that combines vortex and circulation data. Precise background statements and more formal details are collected in Appendix C.

3.1 Complex structure and harmonic 1-forms

Let M be a closed, oriented smooth surface. A (smooth) *almost complex structure* on M is a bundle map

$$J : TM \rightarrow TM, \quad J^2 = -\text{id},$$

whose $\pm i$ -eigenspaces in $TM \otimes \mathbb{C}$ have complex rank one. We call J *integrable* if it comes from a holomorphic atlas; on a surface, integrability is equivalent to the Newlander–Nirenberg condition and makes (M, J) into a compact Riemann surface.

Given such a J , one can choose a Riemannian metric g compatible with J in the sense that $g(Ju, Jv) = g(u, v)$ and the orientation induced by (g, J) agrees with the given orientation. Fixing such a pair (g, J) , the Hodge star

$$\star : \Omega^1(M) \rightarrow \Omega^1(M)$$

is determined by the Riemannian structure, and on 1-forms it coincides with “rotation by 90° ” via the musical isomorphisms:

$$\star = \flat \circ J \circ \sharp \quad \text{on } \Omega^1(M).$$

Remark 3.1 (Harmonic 1-forms as a complex vector space). *The real vector space of harmonic 1-forms*

$$\mathcal{H}^1(M) := \{\alpha \in \Omega^1(M) \mid d\alpha = 0, \delta\alpha = 0\}$$

has dimension $2g$. Since $\star^2 = -\text{id}$ on $\mathcal{H}^1(M)$, we can declare

$$i \cdot \alpha := \star \alpha$$

and thereby view $\mathcal{H}^1(M)$ as a g -dimensional complex vector space.

On a Riemann surface, the canonical bundle $K := \Lambda^{1,0} T^* M$ has space of holomorphic sections $H^0(M, K)$, which can be identified with the Dolbeault cohomology group $H_{\bar{\partial}}^{1,0}(M)$. This space has complex dimension g , and each holomorphic 1-form $\omega_k \in H^0(M, K)$ is a complex-valued 1-form whose real and imaginary parts

$$\text{Re } \omega_k, \text{Im } \omega_k \in \mathcal{H}^1(M)$$

are real harmonic 1-forms. Every element of $\mathcal{H}^1(M)$ arises as the real or imaginary part of some holomorphic 1-form, so choosing a basis $\{\omega_k\}_{k=1}^g$ of $H^0(M, K)$ identifies

$$H^0(M, K) \cong \mathbb{C}^g, \quad \mathcal{H}^1(M) \cong \mathbb{R}^{2g},$$

with the latter being the underlying real vector space of the former. Later we use this dictionary (holomorphic forms $\leftrightarrow$ harmonic forms, $\mathbb{C}^g \cong \mathbb{R}^{2g}$) to express Abel–Jacobi coordinates in the same real period coordinates that appear in our invariant $F(t)$ from Corollary 2.1.

3.2 Jacobian and Abel–Jacobi map

Definition 3.1 (Jacobian). Let $\{\omega_1, \dots, \omega_g\}$ be a basis of holomorphic 1-forms on M , and let

$$\{a_1, \dots, a_g, b_1, \dots, b_g\}$$

be a symplectic basis of $H_1(M; \mathbb{Z})$, i.e. a basis such that the intersection pairing satisfies

$$a_i \cdot a_j = 0, \quad b_i \cdot b_j = 0, \quad a_i \cdot b_j = \delta_{ij}.$$

By changing the basis of holomorphic 1-forms, we may assume that it is normalized:

$$\int_{a_j} \omega_i = \delta_{ij}.$$

The remaining periods

$$\tau_{ij} := \int_{b_j} \omega_i$$

form a $g \times g$ complex matrix τ with $\tau^T = \tau$ and $\text{Im } \tau > 0$. The period lattice is

$$\Lambda := \{m + \tau n \mid m, n \in \mathbb{Z}^g\} \subset \mathbb{C}^g.$$

The Jacobian variety is the complex torus

$$\text{Jac}(M) := \mathbb{C}^g / \Lambda.$$

Equivalently, Hodge theory identifies $H^1(M; \mathbb{R})$ with the space of harmonic 1-forms $\mathcal{H}^1(M)$ (Remark 3.1), and $H^1(M; \mathbb{Z})$ is exactly the lattice of cohomology classes with integral periods. Taking the quotient

$$\text{Jac}(M) \cong H^1(M; \mathbb{R}) / H^1(M; \mathbb{Z})$$

therefore produces the same real $2g$ -dimensional torus as $\mathbb{C}^g / \Lambda$ in the classical Abel–Jacobi picture.

Definition 3.2 (Divisors and degree). A (complex) divisor on M is a function $D : M \rightarrow \mathbb{Z}$ with finite support, i.e., $D(p) = 0$ for all but finitely many points $p \in M$. Equivalently, there exist points $p_1, \dots, p_N \in M$ and integers $n_1, \dots, n_N \in \mathbb{Z}$ such that

$$D = \sum_{k=1}^N n_k p_k,$$

and we define the degree

$$\deg D := \sum_{k=1}^N n_k.$$

If f is a nonzero meromorphic function on M , its divisor (f) records the orders of zeros ($n_k > 0$) and poles ($n_k < 0$) at each point; such divisors always have degree zero. More generally, a nonzero meromorphic section of a line bundle has a divisor that records its zeros and poles in exactly the same way.

We write $\text{Div}^0(M)$ for the group of degree-zero divisors, and $\text{Prin}(M)$ for the subgroup of principal divisors, i.e., divisors of meromorphic functions.

Fix a basepoint $p_0 \in M$. For a degree-zero divisor $D = \sum_k n_k p_k$ we define:

Definition 3.3 (Abel–Jacobi map). The Abel–Jacobi map is

$$\text{AJ}(D) := \sum_k n_k \left(\int_{p_0}^{p_k} \omega_1, \dots, \int_{p_0}^{p_k} \omega_g \right) \mod \Lambda \in \text{Jac}(M),$$

where each path from p_0 to p_k is chosen arbitrarily. Changing paths changes the integrals by elements of the period lattice Λ , so the class in $\text{Jac}(M)$ is well-defined.

Classically, the Abel–Jacobi map measures exactly the obstruction for a divisor to be principal:

Proposition 3.1 (Principal divisors and the Jacobian). The Abel–Jacobi map induces an isomorphism

$$\text{Div}^0(M) / \text{Prin}(M) \xrightarrow{\cong} \text{Jac}(M).$$

In particular, a degree-zero divisor D satisfies $\text{AJ}(D) = 0$ if and only if there exists a global meromorphic function whose divisor is D . Nonzero points of $\text{Jac}(M)$ therefore measure how far a given pattern of zeros/poles is from being the divisor of a single-valued meromorphic function.

A point-vortex configuration

$$D(t) := \sum_a \Gamma_a D_a(t)$$

is morally a divisor with coefficients Γ_a . In classical algebraic-geometry terms one usually takes $n_k \in \mathbb{Z}$, so for our discussion we will assume $\Gamma_a \in \mathbb{Z}$ when we want to regard the vortex configuration as a genuine divisor. Intuitively, $\text{AJ}(D(t))$ tracks where the vortices and their strengths sit in the Jacobian.

3.3 Line bundles and handle phases

We now recall how degree-zero line bundles and their holonomy data also land in $\text{Jac}(M)$.

A (complex) line bundle $L \rightarrow M$ is a rank-one complex vector bundle. A holomorphic structure on L is specified by holomorphic transition functions; holomorphic sections are those that look holomorphic in all trivializations.

Definition 3.4 (Isomorphism classes and Picard group). Two holomorphic line bundles L and L' on M are isomorphic if there exists a fiberwise complex-linear, holomorphic bijection $\Phi : L \rightarrow L'$ covering the identity on M . The set of isomorphism classes of holomorphic line

bundles, with group law given by tensor product,

$$[L] + [L'] := [L \otimes L'],$$

forms the Picard group $\text{Pic}(M)$.

A basic invariant of a line bundle is its degree:

$$\deg(L) := \langle c_1(L), [M] \rangle \in \mathbb{Z},$$

where $c_1(L) \in H^2(M; \mathbb{Z}) \cong \mathbb{Z}$ is the first Chern class and $[M]$ is the fundamental class. For each integer d we write

$$\text{Pic}^d(M) := \{[L] \in \text{Pic}(M) \mid \deg(L) = d\}.$$

Remark 3.2 (Degree-zero bundles and torsor picture). *Only $\text{Pic}^0(M)$ is a subgroup of $\text{Pic}(M)$: it contains the identity (the trivial bundle) and is closed under tensor products and inverses. Each $\text{Pic}^d(M)$ with $d \neq 0$ is not a group by itself but a torsor (translated copy) for $\text{Pic}^0(M)$: once we pick a reference bundle L_0 of degree d , every other degree- d bundle can be written uniquely as $L_0 \otimes T$ with $T \in \text{Pic}^0(M)$.*

On a Riemann surface, $\text{Pic}^0(M)$ and $\text{Jac}(M)$ are canonically isomorphic:

Proposition 3.2 (Picard–Jacobi correspondence). *There is a canonical isomorphism of complex Lie groups*

$$\text{Pic}^0(M) \cong \text{Jac}(M).$$

Thus each degree-zero holomorphic line bundle corresponds to a unique point of $\text{Jac}(M)$, and tensor product of degree-zero bundles matches addition on the Jacobian.

Remark 3.3. *We will not construct the complex manifold structure on $\text{Pic}^0(M)$ explicitly; for our purposes it is enough to know that $\text{Pic}^0(M)$ carries a natural structure of a complex torus, and that the isomorphism in Proposition 3.2 identifies it with the Jacobian $\text{Jac}(M) = \mathbb{C}^g / \Lambda$. Standard references (e.g. Griffiths–Harris) build this structure by realizing $\text{Pic}^0(M)$ as $\text{Div}^0(M) / \text{Prin}(M)$ and endowing it with the quotient topology and complex structure induced from $\mathbb{C}^g$ via the Abel–Jacobi map.*

Physically, it is convenient to think of degree-zero line bundles as carrying a flat $U(1)$ connection, whose holonomy records phase changes around handles.

Proposition 3.3 (Flat connections and handle phases). *Every holomorphic line bundle carries a Hermitian metric and its associated unitary Chern connection. A degree-zero line bundle L lies in $\text{Pic}^0(M)$ if and only if it admits a flat (zero-curvature) unitary connection ∇ . For such a flat connection, the holonomy around each $\gamma \in H_1(M; \mathbb{Z})$ is a phase in S^1 , and the $2g$ independent holonomy phases (modulo integer relations) determine a point in $\text{Jac}(M)$ that agrees with the class of L under the isomorphism of Proposition 3.2.*

In short:

- divisor data (where zeros and poles sit) defines a point of $\text{Jac}(M)$ via AJ;
- connection data (how phases accumulate around handles) defines a point of $\text{Jac}(M)$ via holonomy;
- both descriptions talk about the same Jacobian.

3.4 Point vortices as meromorphic sections

We now package the point-vortex and harmonic-circulation data into this complex-geometric language.

Recall that in the classical planar story one introduces a complex potential

$$W = \psi + i\phi,$$

whose real part ψ is the stream function and whose imaginary part ϕ is a velocity potential; differentiating W and rotating by 90° recovers the velocity. On a simply connected domain this works globally: W is a single-valued holomorphic function with logarithmic singularities at the vortex locations, and the divisor of W (more precisely, of e^W) encodes the vortex strengths. The global compatibility condition on total circulation matches the degree-zero condition on divisors.

On a higher-genus closed surface, two things break:

- the velocity 1-form has a harmonic part h that cannot be written as $\star d\psi$;
- On a multiply connected surface, analytic continuation of the complex potential W around a nontrivial homology loop γ can change it by an additive period

$$W \mapsto W + P_\gamma, \quad P_\gamma \in \mathbb{C},$$

so W is only defined up to such periods and is therefore not a globally single-valued function. (basically, its exponential e^W picks up a multiplicative phase factor).

A convenient way to encode both phenomena is to treat the complex potential as a *local* logarithm of a meromorphic section of a line bundle.

Definition 3.5 (Vortex line bundle (formal picture)). *Formally, associate to a vortex configuration a holomorphic line bundle $L \rightarrow M$ equipped with:*

- a meromorphic section s whose divisor is the vortex divisor
- $$(s) = D(t) := \sum_a \Gamma_a D_a(t);$$
- a unitary connection ∇ which encodes the harmonic handle circulations.

In a local trivialization of L away from the vortices, we can write $s = \exp(W)$ for a holomorphic function $W = \psi + i\phi$, defined up to additive integer multiples of $2\pi i$ (from going around zeros/poles) and up to additive periods from parallel transport along handles. The divisor (s) records the vortex strengths, while the monodromy of s around handles records the harmonic circulations.

Under the Picard–Jacobi identification, the isomorphism class $[L] \in \text{Pic}^0(M)$ corresponds to a point of $\text{Jac}(M)$ which can be decomposed into:

- a contribution from the divisor $D(t)$, given by $\text{AJ}(D(t))$;
- a contribution from the flat part of the connection, given by its handle holonomy phases.

The key observation is that these contributions *add* in $\text{Jac}(M)$: tensoring a “pure-vortex” bundle with trivial holonomy and a “pure-holonomy” bundle with trivial divisor simply adds their Jacobian points.

We treat this bundle-and-connection picture as a guiding formalism; a fully rigorous construction is still work in progress.

3.5 A Jacobian-valued invariant

We now revisit the real invariant from Corollary 2.1 in this complex-analytic language. Recall that, for an L^2 -orthonormal basis $\{h_i\}_{i=1}^{2g}$ of harmonic 1-forms,

$$c_i(t) := \int_M \eta(t) \wedge \star h_i$$

are the harmonic circulation coordinates of the flow, and that Lemma 2.4 shows formally that there exist local scalar potentials A_i with $\star h_i = dA_i$ near each vortex such that

$$F_i(t) := c_i(t) - \sum_a \Gamma_a A_i(D_a(t))$$

is constant in time. Collecting these into $F(t) \in \mathbb{R}^{2g}$ gives the invariant of Corollary 2.1.

The complex structure lets us reinterpret $F(t)$ as coordinates of a point in $\text{Jac}(M)$: roughly, the A_i function as real period coordinates for holomorphic 1-forms, while the c_i are real harmonic circulation coordinates. Identifying these two pieces is a matter of choosing the right basis of holomorphic forms and the corresponding real/imaginary parts; see Appendix C.4 for more detail.

The following theorem is the main conceptual statement of this section. Its proof is formal and relies on the identifications sketched above; a fully rigorous complex-analytic proof is work in progress.

Theorem 3.1 (Jacobian-valued invariant for point vortices (formal)). *Equip M with a complex structure compatible with its orientation, turning it into a compact Riemann surface. Let $\{\omega_k\}_{k=1}^g$ be a basis of holomorphic 1-forms, and let*

$$D(t) := \sum_a \Gamma_a D_a(t)$$

be the vortex divisor, where each point vortex $D_a(t) \in M$ is advected by the flow and has strength $\Gamma_a \in \mathbb{Z}$. Let $\text{AJ}(D(t)) \in \text{Jac}(M)$ denote the Abel–Jacobi image of $D(t)$.

Let $c(t) \in \mathbb{R}^{2g}$ be the harmonic circulation coordinates of the flow, defined using an L^2 -orthonormal basis of harmonic 1-forms as above. Then there exists a linear map on the universal cover

$$\tilde{H} : \mathbb{R}^{2g} \rightarrow \mathbb{R}^{2g}$$

whose projection to $\mathbb{R}^{2g}/\Lambda \cong \text{Jac}(M)$ defines a smooth group homomorphism

$$H : \mathbb{R}^{2g} \rightarrow \text{Jac}(M),$$

such that the combination

$$J(t) := \text{AJ}(D(t)) + H(c(t)) \in \text{Jac}(M)$$

is constant in time, i.e.,

$$\frac{d}{dt} J(t) = 0 \quad \text{along the point-vortex dynamics.}$$

Interpreted via Proposition 3.2, $J(t)$ is the isomorphism class of a degree-zero holomorphic line bundle $L(t) \in \text{Pic}^0(M)$ that combines the vortex divisor and handle circulations. The Euler evolution advects the vortices and deforms the metric, but the class $[L(t)] \in \text{Pic}^0(M)$ remains fixed.

Remark 3.4 (Interpretation and status). *The theorem says that, although the Euler flow is a real, area-preserving dynamics that does not preserve the complex structure in general, the combined data of*

vortex positions and harmonic circulations defines a holomorphic line bundle whose isomorphism class is conserved. Equivalently, all point-vortex configurations reachable from a given initial condition define holomorphic line bundles lying in the same component $\text{Pic}^0(M)$ and in the same isomorphism class.

At the level of differential forms, this is the complex-analytic shadow of the real invariant $F(t)$ in Corollary 2.1. Making this correspondence fully rigorous requires a careful construction of the “vortex line bundle” and its Chern connection, and a precise matching between the real period coordinates used for $F(t)$ and the Abel–Jacobi coordinates on $\text{Jac}(M)$; these steps are only outlined here and are part of the ongoing project. A more systematic discussion is given in Appendix C.4.

A Supplementary Background for Introduction Section

A.1 Classical planar point-vortex formulation

For completeness, we recall the standard complex-analytic formulation of point vortices in the plane. Let $\Omega = \sum_{a=1}^N \Gamma_a \delta_{z_a}$ be a vorticity field concentrated at points $z_a \in \mathbb{C}$ with strengths Γ_a . The scalar stream function ψ solves

$$\Delta \psi = \sum_{a=1}^N \Gamma_a \delta_{z_a},$$

and the incompressible velocity is recovered as $v = \nabla^\perp \psi$, i.e., a 90° rotation of $\nabla \psi$. On simply connected domains, one can package the stream function ψ and a velocity potential ϕ into a complex potential

$$W(z) = \psi(z) + i\phi(z),$$

which is holomorphic away from the vortex locations. A single vortex of strength Γ at z_0 corresponds to a logarithmic singularity

$$W(z) \sim \frac{\Gamma}{2\pi i} \log(z - z_0),$$

and a general configuration is obtained by summing such contributions, plus possibly a polynomial part representing background flow. On the Riemann sphere or other compactifications, the divisor of W must have total degree zero, so the vortex strengths must satisfy a global compatibility condition (heuristically, “total circulation plus circulation at infinity = 0”). This is the planar precursor of the degree-zero divisor and Abel–Jacobi constraints that appear on closed Riemann surfaces.

A.2 Topological degrees of freedom and DEC-based solvers

In the main text we emphasized that most vortex methods and many DEC-based fluid solvers expose only part of the topological structure. Here we briefly justify this statement.

Classical vortex methods approximate vorticity by particles or blobs and evolve them via Biot–Savart integrals or equivalent formulations [Cottet and Koumoutsakos 2000; Chorin 1994; Leonard 1980; Majda and Bertozzi 2001]. When these methods are applied to simply connected planar domains (e.g., the plane or a disk), the main global constraint coming from topology is total circulation: the first Betti number vanishes, so there are no nontrivial handle circulations or harmonic 1-forms to track. On multiply connected or curved domains, the same formulations can in principle be adapted via suitable Green’s functions and image systems, but the handle circulations then live implicitly inside renormalized energies and

boundary-image constructions, rather than as explicit degrees of freedom.

DEC-based solvers such as simplicial fluids [Elcott et al. 2007] and DEC formulations of surface Navier–Stokes [Desbrun et al. 2008; Camporesi and Kanso 2016] start from a discrete Hodge decomposition

$$C^1(K) = \text{im}(d_0) \oplus \text{im}(\delta_2) \oplus \mathcal{H}_h^1(K),$$

where $\mathcal{H}_h^1(K)$ is the discrete harmonic subspace. In practice, these methods update the full discrete velocity or vorticity field and typically either work on topologically simple meshes or treat the harmonic subspace as part of a nullspace that is fixed implicitly by boundary conditions or initial data. The recent Fluid Cohomology framework [Yin et al. 2023b,a] systematically isolates these harmonic components, derives their equations of motion, and shows that they form a finite-dimensional cohomological space coupled to the usual PDE variables.

Our point-vortex formulation can be viewed as a reduced model in this spirit: instead of evolving a full velocity field on the mesh, we evolve a finite set of vortex positions together with a finite-dimensional vector of harmonic coordinates. By building this reduced model on the same DEC-based Hodge decomposition, we inherit the cohomological structure exposed by Fluid Cohomology, now realized exactly at the level of point-vortex degrees of freedom.

B DEC AND HODGE-THEORETIC DETAILS

In this appendix we collect the more technical details behind the DEC and Hodge-theoretic statements in Section 2. Nothing here changes the main narrative; the goal is simply to justify the formal manipulations used in the main text and to make precise what is meant by “projection onto the harmonic subspace” and the evolution of harmonic coefficients.

B.1 Exterior-calculus form of Euler

We briefly justify Proposition 2.1. Let $\eta = v^\flat$ be the metric dual of v and recall that for a function f and 1-form α on a Riemannian manifold we have:

- $\sharp : T^*M \rightarrow TM$ and $\flat : TM \rightarrow T^*M$ are metric duality isomorphisms;
- the divergence of v can be written as $\delta\eta = -\star d\star\eta$;
- the vorticity 2-form is $\Omega := d\eta$.

The incompressibility condition $\nabla \cdot v = 0$ becomes

$$0 = \nabla \cdot v = \delta\eta.$$

To rewrite the momentum equation, start from

$$\partial_t v + (v \cdot \nabla)v = -\nabla p.$$

Apply $\flat$ to obtain

$$\partial_t \eta + (v \cdot \nabla)\eta^\flat = -d p.$$

Using the Cartan identity and basic identities on 1-forms (see, e.g., [Arnold and Khesin 1998]), one can show that

$$(v \cdot \nabla)v^\flat = \iota_v d\eta + \frac{1}{2}d(\|v\|^2).$$

Thus, up to adding a gradient to the pressure, the momentum equation becomes

$$\partial_t \eta + \iota_v \Omega + d\tilde{p} = 0$$

for some scalar field $\tilde{p}$. This is exactly (3). Since any two choices of pressure that differ by a function of time only are equivalent physically, we can simply call $\tilde{p}$ the pressure and obtain Proposition 2.1.

B.2 Hodge decomposition and incompressible 1-forms

A standard Hodge-theory result on a closed Riemannian manifold (M, g) of dimension n says that every k -form α admits an orthogonal decomposition

$$\alpha = d\beta \oplus \delta\gamma \oplus h,$$

where h is harmonic, $dh = 0$ and $\delta h = 0$, and

$$\mathcal{H}^k(M) \cong H^k(M; \mathbb{R})$$

is finite-dimensional and independent of g up to isomorphism. For a surface ($n = 2$) and $k = 1$, this specializes to Proposition 2.2:

$$\alpha = d\phi + \star d\psi + h.$$

On a closed surface, harmonic scalar functions are constant. Indeed, if $\Delta\phi = 0$ and M is compact, then

$$0 = \int_M \phi \Delta\phi \, dA = - \int_M \|\nabla\phi\|^2 \, dA$$

shows $\nabla\phi = 0$ and hence ϕ is constant. This observation is what kills the gradient part in the incompressible case.

B.3 Stream function, harmonic part, and Poisson problem

We now justify Lemmas 2.1 and 2.2.

Starting from the Hodge decomposition

$$\eta = d\phi + \star d\psi + h,$$

apply the codifferential:

$$\delta\eta = \delta d\phi + \delta(\star d\psi) + \delta h.$$

On 1-forms in two dimensions, $\delta(\star d\psi) = 0$ because

$$\delta(\star d\psi) = -\star d\star(\star d\psi) = -\star d^2\psi = 0,$$

and by definition $\delta h = 0$ for harmonic h . Thus

$$\delta\eta = \delta d\phi = \Delta\phi.$$

If η is incompressible, $\delta\eta = 0$ and we get $\Delta\phi = 0$. As discussed above, on a closed surface this implies ϕ is constant, hence $d\phi = 0$ and we obtain Lemma 2.1:

$$\eta = \star d\psi + h.$$

For Lemma 2.2, note that the vorticity 2-form is

$$\Omega = d\eta = d(\star d\psi),$$

since $dh = 0$. Using $d\star d\psi = (\Delta\psi) \, dA$ in two dimensions, we get

$$\Omega = (\Delta\psi) \, dA,$$

so the scalar vorticity is $\omega = \Delta\psi$. On a closed surface, the Laplacian has constants as kernel and its image consists of mean-zero functions. The compatibility condition

$$\int_M \omega \, dA = 0$$

is exactly the requirement that ω lie in the image of Δ . When this holds, there exists a solution ψ unique up to an additive constant, giving Lemma 2.2. If the mean of ω is nonzero, one must subtract a

constant background (or equivalently work with a rigid rotation) to restore compatibility.

B.4 Discrete harmonic basis construction

We sketch the construction behind Lemma 2.3. Let K be a triangulated closed genus- g surface.

Step 1: Tree-cotree decomposition and cycle generators. Choose:

- a primal spanning tree T in the 1-skeleton K^1 ;
- a dual spanning tree T^* in the dual graph (vertices are faces, edges cross primal edges).

Edges that are in neither T nor T^* form a set E_{cycle} of $2g$ edges. Each such edge closes a fundamental cycle in the primal mesh. For each $e \in E_{\text{cycle}}$, construct a 1-cochain $\alpha_e \in C^1(K)$ that is 1 on e , zero on other edges, and extended along the unique path in T needed to make it represent the corresponding cycle. The set $\{\alpha_e\}$ spans a $2g$ -dimensional subspace that maps onto $H_1(M; \mathbb{R})$.

Step 2: Projection onto the harmonic subspace. The discrete 1-form Laplacian is

$$\Delta_1 := \delta_2 d_1 + d_0 \delta_1.$$

On a closed mesh, $\ker \Delta_1$ is the discrete harmonic subspace:

$$\ker \Delta_1 = \ker d_1 \cap \ker \delta_1 = \mathcal{H}_h^1(K).$$

For each initial cycle cochain α_e , we compute its orthogonal projection onto $\ker \Delta_1$ with respect to the discrete L^2 inner product.

Concretely, we solve

$$\Delta_1 x = \Delta_1 \alpha_e$$

for $x \in C^1(K)$ in the least-squares sense (we can, for example, fix the nullspace by enforcing orthogonality to a precomputed basis of $\ker \Delta_1$). Then

$$\eta_e := \alpha_e - x$$

satisfies $\Delta_1 \eta_e = 0$, hence $d_1 \eta_e = 0$ and $\delta_1 \eta_e = 0$. The set $\{\eta_e\}_{e \in E_{\text{cycle}}}$ spans $\mathcal{H}_h^1(K)$.

Step 3: Orthonormalization. Finally, apply Gram-Schmidt with respect to the L^2 inner product

$$\langle \alpha, \beta \rangle_{L^2} = \alpha^\top \star_1 \beta$$

to obtain an L^2 -orthonormal basis $\{\eta_a\}_{a=1}^{2g}$. This finishes the construction in Lemma 2.3.

In practice, one can reuse standard code for discrete harmonic form computation as used in, e.g., Desbrun et al. [2008].

B.5 Evolution of harmonic coefficients (formal calculation)

We expand on the formal steps leading to Lemma 2.4. Recall

$$c_i(t) = \int_M \eta(t) \wedge \star h_i.$$

Differentiate in time:

$$\dot{c}_i(t) = \frac{d}{dt} \int_M \eta \wedge \star h_i = \int_M \partial_t \eta \wedge \star h_i.$$

Insert the 1-form form of Euler (3):

$$\partial_t \eta = -\iota_v \Omega - dp.$$

Thus

$$\dot{c}_i(t) = - \int_M \iota_v \Omega \wedge \star h_i - \int_M dp \wedge \star h_i.$$

The pressure term vanishes because h_i is harmonic:

$$\int_M dp \wedge \star h_i = \langle dp, h_i \rangle_{L^2} = \langle p, \delta h_i \rangle_{L^2} = 0.$$

We are left with

$$\dot{c}_i(t) = - \int_M \iota_v \Omega \wedge \star h_i.$$

Now insert the point-vortex vorticity

$$\Omega = \sum_a \Gamma_a \delta_{D_a(t)}.$$

Formally,

$$\dot{c}_i(t) = - \sum_a \Gamma_a \int_M \iota_v \delta_{D_a(t)} \wedge \star h_i.$$

The Dirac 2-current $\delta_{D_a(t)}$ acts on 0-forms by evaluation at $D_a(t)$, and on 1-forms by pairing with vectors at $D_a(t)$. In a small coordinate disk around $D_a(t)$, we can write

$$\star h_i = dA_i$$

for some scalar potential A_i (unique up to an additive constant and up to periods if we move around handles).

Interpreting the pairing in a weak sense, we have

$$\int_M \iota_v \delta_{D_a(t)} \wedge dA_i \sim \nabla A_i(D_a(t)) \cdot \dot{D}_a(t) = \frac{d}{dt} A_i(D_a(t)),$$

where $\dot{D}_a(t) = v(D_a(t))$ is the vortex velocity. Summing over vortices gives

$$\dot{c}_i(t) = \frac{d}{dt} \left(\sum_a \Gamma_a A_i(D_a(t)) \right),$$

which is (12). Thus

$$F_i(t) = c_i(t) - \sum_a \Gamma_a A_i(D_a(t))$$

is constant in time.

A fully rigorous treatment would use the language of currents and carefully justify the manipulations with $\iota_v \delta_{D_a(t)}$; here we only need the formal structure of the evolution law.

B.6 Discrete evolution of harmonic coordinates

We now describe the evolution of the harmonic coordinates in the actual implementation and how it relates to the continuous formula from Lemma 2.4.

From potentials A_i to a velocity-based formula. Lemma 2.4 says, formally,

$$\dot{c}_i(t) = \frac{d}{dt} \left(\sum_a \Gamma_a A_i(D_a(t)) \right),$$

where $\star h_i = dA_i$ near each vortex. Using the point-vortex ODE $\dot{D}_a(t) = v(D_a(t))$ and the chain rule,

$$\frac{d}{dt} A_i(D_a(t)) = \nabla A_i(D_a(t)) \cdot \dot{D}_a(t) = \nabla A_i(D_a(t)) \cdot v(D_a(t)).$$

On a surface, the gradient of A_i can be expressed in terms of the harmonic 1-form h_i and the 90° rotation J . Indeed,

$$\star h_i = dA_i \implies \nabla A_i = (dA_i)^\# = (\star h_i)^\#.$$

Using the metric-duality identity $\star = \flat \circ J \circ \sharp$ on 1-forms, we get

$$(\star h_i)^\sharp = J h_i^\sharp.$$

Thus the evolution law can be written purely in terms of vector fields:

$$\dot{c}_i(t) = \sum_a \Gamma_a v(D_a(t)) \cdot J h_i^\sharp(D_a(t)). \quad (15)$$

This is exactly the continuous ODE we discretize in practice.

Discrete fields and interpolation. On the mesh, we have:

- A discrete stream function $\psi_h \in C^0(K)$ solving the Poisson system (9).
- A discrete harmonic basis $\{\eta_a\}_{a=1}^{2g} \subset C^1(K)$ from Lemma 2.3.
- A discrete velocity 1-cochain

$$u_h(t) = \star_1 d_0 \psi_h(t) + h_h(t), \quad h_h(t) = \sum_a c_a(t) \eta_a.$$

To evaluate the right-hand side of (15), we need vector fields on triangles. We use a standard DEC-to-vector reconstruction: on each triangle we interpret a 1-cochain (edge-integrated form) as a piecewise linear vector field that has the prescribed circulations along edges and is as uniform as possible inside the face. Applying this reconstruction to u_h and to each harmonic basis cochain η_a yields piecewise linear vector fields

$$v_h(t), \quad h_a^\sharp$$

defined per face. The 90° rotation J is applied per face using the local metric (or embedding) to rotate tangent vectors.

Given a vortex at position $D_k(t)$ inside some triangle, we evaluate:

- The discrete velocity $v_h(D_k(t))$ by barycentric interpolation of the per-face vector field.
- The harmonic vector fields $h_a^\sharp(D_k(t))$ and their rotated versions $J h_a^\sharp(D_k(t))$ similarly.

ODE system for $c_a(t)$ and time integration. We now discretize (15) by replacing continuous quantities with their DEC-based approximations:

$$\frac{dc_a}{dt} \approx \sum_k \Gamma_k v_h(D_k(t)) \cdot J h_a^\sharp(D_k(t)), \quad a = 1, \dots, 2g.$$

This gives a coupled ODE system for

$$\{D_k(t)\}_{k=1}^N \quad \text{and} \quad c(t) = (c_1(t), \dots, c_{2g}(t)),$$

where the vortex positions evolve by the discrete point-vortex rule

$$\dot{D}_k(t) = v_h(D_k(t))$$

and the harmonic coordinates evolve by the ODE above.

In the implementation, we simply integrate this coupled ODE system using a standard explicit Runge-Kutta scheme (RK4), recomputing ψ_h from the current discrete vorticity and updating u_h at each substep. The real invariant $F(t)$ from Corollary 2.1 is then preserved up to the usual time-discretization error: at small time steps we observe that $F(t)$ remains nearly constant, with deviations controlled by the RK4 integration error rather than by the spatial discretization.

Interpretation. Conceptually, this approach should be viewed as a “particle + finite-element” discretization of the continuous evolution law:

- DEC provides a geometry- and topology-aware approximation of the underlying fields $(v, h_i^\sharp, J)$.
- Particles (point vortices) sample these fields at their positions.
- The ODE (15) ties the evolution of the harmonic coordinates $c_i(t)$ directly to these samples.

C Complex-analytic background and formal details

This appendix collects background material and sketches that underlie the complex-analytic side of Section 3. We keep statements at the level needed in the main text and refer to standard texts on Riemann surfaces and algebraic geometry for proofs.

C.1 Harmonic forms and holomorphic forms

Let M be a compact Riemann surface of genus g . Denote by K the canonical bundle of M and by $H^0(M, K)$ the space of holomorphic 1-forms, which has complex dimension g . The space of harmonic 1-forms

$$\mathcal{H}^1(M) = \{\alpha \in \Omega^1(M) \mid d\alpha = 0, \delta\alpha = 0\}$$

has real dimension $2g$ and is canonically isomorphic to $H^1(M; \mathbb{R})$ via Hodge theory.

Each holomorphic 1-form ω can be written as

$$\omega = \alpha + i\beta$$

with $\alpha, \beta \in \mathcal{H}^1(M)$; conversely, every harmonic 1-form arises as a real or imaginary part of a holomorphic 1-form. Choosing a complex basis $\{\omega_k\}_{k=1}^g$ of $H^0(M, K) \cong \mathbb{C}^g$ therefore yields a real basis

$$\{\operatorname{Re} \omega_1, \operatorname{Im} \omega_1, \dots, \operatorname{Re} \omega_g, \operatorname{Im} \omega_g\}$$

of $\mathcal{H}^1(M) \cong \mathbb{R}^{2g}$. This is the identification used implicitly when we regard $\operatorname{Jac}(M)$ both as $\mathbb{C}^g/\Lambda$ and as $H^1(M; \mathbb{R})/H^1(M; \mathbb{Z})$.

C.2 Jacobian, divisors, and Picard

For completeness, we record the standard results used in Section 3.

Proposition C.1 (Divisors modulo principal divisors). *Let $\operatorname{Div}^0(M)$ be the group of degree-zero divisors and $\operatorname{Prin}(M)$ the subgroup of principal divisors. Then the Abel–Jacobi map induces an isomorphism*

$$\operatorname{Div}^0(M)/\operatorname{Prin}(M) \xrightarrow{\cong} \operatorname{Jac}(M).$$

Proposition C.2 (Picard group and degree-zero component). *The Picard group $\operatorname{Pic}(M)$ of a compact Riemann surface M decomposes as*

$$\operatorname{Pic}(M) = \bigsqcup_{d \in \mathbb{Z}} \operatorname{Pic}^d(M),$$

where $\operatorname{Pic}^d(M)$ denotes isomorphism classes of holomorphic line bundles of degree d . The degree-zero component $\operatorname{Pic}^0(M)$ is a complex Lie group and there is a canonical isomorphism

$$\operatorname{Pic}^0(M) \cong \operatorname{Jac}(M).$$

The proofs proceed by constructing, for each degree-zero divisor D , a line bundle $\mathcal{O}(D)$ with a meromorphic section whose divisor is D , and then identifying $\operatorname{Div}^0(M)/\operatorname{Prin}(M)$ with $\operatorname{Pic}^0(M)$ by sending D to $[\mathcal{O}(D)]$. Composing with the isomorphism of Proposition C.1 yields Proposition C.2.

C.3 Flat line bundles and holonomy

We recall the relation between degree, curvature, and holonomy for line bundles.

Proposition C.3 (Chern–Weil theory for line bundles). *Let $L \rightarrow M$ be a holomorphic line bundle with Hermitian metric and Chern connection ∇^{Ch} . Its curvature $F_{\nabla^{\text{Ch}}}$ is a purely imaginary 2-form, and*

$$\deg(L) = \frac{i}{2\pi} \int_M F_{\nabla^{\text{Ch}}}.$$

In particular, L has degree zero if and only if the curvature of its Chern connection is exact with vanishing integral, and after adjusting the metric one can arrange for a flat unitary connection.

On a compact Riemann surface, a flat unitary line bundle is uniquely determined (up to isomorphism) by its holonomy around a basis of $H_1(M; \mathbb{Z})$. The map “holonomy modulo integer relations” realises the isomorphism $\text{Pic}^0(M) \cong \text{Jac}(M)$ used in Proposition 3.3.

C.4 Formal derivation of the Jacobian-valued invariant

We briefly outline how the real invariant of Corollary 2.1 can be viewed as coordinates of a fixed Jacobian point.

Let $\{h_i\}_{i=1}^{2g}$ be an L^2 -orthonormal basis of harmonic 1-forms and let $c_i(t)$ be defined by

$$c_i(t) := \int_M \eta(t) \wedge \star h_i,$$

with $\eta = v^\flat$ the velocity 1-form. Lemma 2.4 shows formally that there exist local potentials A_i with $\star h_i = dA_i$ near each vortex such that

$$F_i(t) := c_i(t) - \sum_a \Gamma_a A_i(D_a(t))$$

is constant in time. Collecting these into

$$F(t) := (F_1(t), \dots, F_{2g}(t)) \in \mathbb{R}^{2g}$$

gives the invariant of Corollary 2.1.

To relate $F(t)$ to $\text{Jac}(M)$, choose a basis of holomorphic forms $\{\omega_k\}_{k=1}^g$ and write

$$\omega_k = \alpha_k + i\beta_k, \quad \alpha_k, \beta_k \in \mathcal{H}^1(M),$$

with $\{\alpha_1, \beta_1, \dots, \alpha_g, \beta_g\}$ an L^2 -orthonormal basis of $\mathcal{H}^1(M)$. In this basis, the Abel–Jacobi map of the vortex divisor becomes

$$\text{AJ}(D(t)) = (U_1(t) + iV_1(t), \dots, U_g(t) + iV_g(t)) \mod \Lambda,$$

where U_k, V_k are (up to choice of basepoint and periods) real-valued functions built from integrating α_k, β_k along paths to the vortex positions.

The harmonic circulation coordinates $c_i(t)$ can similarly be expressed in this basis and collected into a vector $c(t) \in \mathbb{R}^{2g}$. The potentials A_i in Lemma 2.4 can be chosen compatibly with the period coordinates associated to ω_k , so that the constants $F_i(t)$ correspond exactly to the components of a fixed vector in $\mathbb{R}^{2g}$, up to the period lattice Λ .

The linear map $\tilde{H} : \mathbb{R}^{2g} \rightarrow \mathbb{R}^{2g}$ in Theorem 3.1 is then obtained by expressing the harmonic basis h_i in terms of $\{\alpha_k, \beta_k\}$, and H is the induced map on the torus $\mathbb{R}^{2g}/\Lambda \cong \text{Jac}(M)$. In these coordinates, the constancy of $F(t)$ is equivalent to the constancy of

$$J(t) := \text{AJ}(D(t)) + H(c(t)) \in \text{Jac}(M),$$

which is exactly the content of Theorem 3.1.

A fully rigorous proof would require:

- carefully constructing the vortex line bundle and its Chern connection so that curvature matches vorticity and holonomy matches harmonic circulations;
- checking that the resulting line bundle lies in $\text{Pic}^0(M)$ and that its class is indeed $J(t)$.

These analytic and algebro-geometric details lie beyond the scope of this note and are part of the ongoing research project; for the purposes of this report we treat Theorem 3.1 as a formal complex-analytic upgrade of the real invariant in Corollary 2.1.

References

- Hassan Aref. 2007. Point vortex dynamics: A classical mathematics playground. *J. Math. Phys.* 48, 6 (2007), 065401. doi:10.1063/1.2738608
- Vladimir I. Arnold and Boris A. Khesin. 1998. *Topological Methods in Hydrodynamics*. Applied Mathematical Sciences, Vol. 125. Springer, New York. doi:10.1007/978-1-4612-1142-2
- Stefanella Boatto and Jair Koiller. 2008. Vortices on closed surfaces. arXiv:0802.4313 [math.SG].
- Stefanella Boatto and Jair Koiller. 2015. Vortices on closed surfaces. In *Geometry, Mechanics, and Dynamics: The Legacy of Jerry Marsden*, D.E. Chang, D.D. Holm, G. Patrick, and T. Ratiu (Eds.). Fields Institute Communications, Vol. 73. Springer, New York, 185–237.
- Guido Boffetta and Robert E. Ecke. 2012. Two-dimensional turbulence. *Annual Review of Fluid Mechanics* 44 (2012), 427–451. doi:10.1146/annurev-fluid-120710-101240
- V. A. Bogomolov. 1977. Dynamics of vorticity on a sphere. *Fluid Dynamics* 12, 6 (1977), 863–870. doi:10.1007/BF01090320
- Francesca Camporesi and Eva Kanso. 2016. Discrete Exterior Calculus for the surface Navier–Stokes equations. arXiv preprint arXiv:1611.04392 (2016).
- Alexandre J. Chorin. 1994. *Vorticity and Turbulence*. Springer-Verlag, New York.
- Georges-Henri Cottet and Petros D. Koumoutsakos. 2000. *Vortex Methods: Theory and Practice*. Cambridge University Press, Cambridge.
- Darren G. Crowdy. 2005. The Schwarz–Christoffel mapping to bounded multiply connected polygonal domains. *Proceedings of the Royal Society A* 461, 2055 (2005), 2653–2678. doi:10.1098/rspa.2005.1498
- Darren G. Crowdy. 2007. Schwarz–Christoffel mappings to unbounded multiply connected polygonal regions. *Mathematical Proceedings of the Cambridge Philosophical Society* 142, 2 (2007), 319–339. doi:10.1017/S0305004106009997
- Darren G. Crowdy and J. Steven Marshall. 2005. The motion of a point vortex around multiple circular islands. *Physics of Fluids* 17, 5 (2005), 056602. doi:10.1063/1.1884612
- Mathieu Desbrun, Eva Kanso, and Yiyong Tong. 2008. Discrete Differential Forms for Computational Modeling. In *ACM SIGGRAPH 2008 Courses*. No. 5, 1–50 (course notes).
- David G. Dritschel and Stefanella Boatto. 2015. The motion of point vortices on closed surfaces. *Proceedings of the Royal Society A* 471, 2182 (2015), 20140890. doi:10.1098/rspa.2014.0890
- Stanley Elcott, Yiyong Tong, Eva Kanso, Peter Schröder, and Mathieu Desbrun. 2007. Stable, circulation-preserving, simplicial fluids. *ACM Transactions on Graphics* 26, 1 (2007), 4:1–4:11. doi:10.1145/1189762.1189766
- Andrew Gillette. 2009. Notes on Discrete Exterior Calculus. Lecture notes. Unpublished notes; commonly cited DEC primer.
- Clodoaldo Grotta-Ragazzo. 2017. Hydrodynamic vortex on surfaces. *Journal of Nonlinear Science* 27, 6 (2017), 1609–1640.
- Clodoaldo Grotta-Ragazzo, Björn Gustafsson, and Jair Koiller. 2024. On the interplay between vortices and harmonic flows: Hodge decomposition of Euler’s equations in 2D. *Regular and Chaotic Dynamics* 29, 2 (2024), 241–303. doi:10.1134/S1560354724020011
- Björn Gustafsson. 2019. Vortex motion and geometric function theory: the role of connections. *Philosophical Transactions of the Royal Society A* 377, 2158 (2019), 20180341. doi:10.1098/rsta.2018.0341
- Björn Gustafsson. 2022. Vortex Pairs and Dipoles on Closed Surfaces. *Journal of Nonlinear Science* 32, 6 (2022), 62. doi:10.1007/s00332-022-09822-9
- Hermann von Helmholtz. 1858. Über Integrale der hydrodynamischen Gleichungen, welche den Wirbelbewegungen entsprechen. *Journal für die reine und angewandte Mathematik* 55 (1858), 25–55.
- Anil N. Hirani. 2003. *Discrete Exterior Calculus*. Ph. D. Dissertation. California Institute of Technology.
- Ravi Kidambi and Paul K. Newton. 1998. Motion of three point vortices on a sphere. *Physica D* 116, 3–4 (1998), 143–175. doi:10.1016/S0167-2789(98)00113-8
- Yoshifumi Kimura. 1987. Similarity solution of two-dimensional point vortices. *Journal of the Physical Society of Japan* 56, 7 (1987), 2024–2030. doi:10.1143/JPSJ.56.2024

- Yoshifumi Kimura. 1999. Vortex motion on surfaces with constant curvature. *Proceedings of the Royal Society A* 455 (1999), 245–259. doi:10.1098/rspa.1999.0311
- Yoshifumi Kimura and Hisashi Okamoto. 1987. Vortex motion on a sphere. *Journal of the Physical Society of Japan* 56, 12 (1987), 4203–4206. doi:10.1143/JPSJ.56.4203
- Gustav Kirchhoff. 1876. *Vorlesungen über mathematische Physik, Band 1: Mechanik*. B.G. Teubner, Leipzig.
- Anthony Leonard. 1980. Vortex methods for flow simulation. *J. Comput. Phys.* 37, 3 (1980), 289–335. doi:10.1016/0021-9991(80)90034-3
- Chia-Chiao Lin. 1941. On the motion of vortices in two dimensions. I. Existence of the Kirchhoff-Routh function. II. Integrals of motion for a vortex system. *Proceedings of the National Academy of Sciences USA* 27, 12 (1941), 570–575, 575–577.
- Andrew J. Majda and Andrea L. Bertozzi. 2001. *Vorticity and Incompressible Flow*. Cambridge University Press, Cambridge.
- Paul K. Newton. 2001. *The N-Vortex Problem: Analytical Techniques*. Springer-Verlag, New York.
- Lorenzo M. Polvani and David G. Dritschel. 1993. Wave and vortex dynamics on the surface of a sphere. *Journal of Fluid Mechanics* 255 (1993), 35–64. doi:10.1017/S0022112093002381
- Constantine Pozrikidis. 2011. A boundary-integral algorithm for adaptive motion planning. *Engineering Analysis with Boundary Elements* 35, 1 (2011), 123–128. doi:10.1016/jenganabound.2010.07.003
- Philip G. Saffman. 1992. *Vortex Dynamics*. Cambridge University Press, Cambridge.
- Takashi Sakajo and Yuuki Shimizu. 2016. Point vortex interactions on a toroidal surface. *Proceedings of the Royal Society A* 472, 2190 (2016), 20160271. doi:10.1098/rspa.2016.0271
- Hang Yin, Mohammad Sina Nabizadeh, Baichuan Wu, and Albert Chern. 2023a. Fluid Cohomology (Technical Report). arXiv:2302.07227.
- Hang Yin, Mohammad Sina Nabizadeh, Baichuan Wu, Stephanie Wang, and Albert Chern. 2023b. Fluid Cohomology. *ACM Transactions on Graphics* 42, 4 (2023), 126:1–126:25. doi:10.1145/3592402